

Leibniz-type inequality: a counterexample

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Problem. Prove or disprove the inequality

$$\|(fg)''\|_2 \leq \|f\|_\infty \|g''\|_2 + \|g\|_\infty \|f''\|_2$$

for every $f, g \in C_0^2(0, \pi)$.

The inequality fails. At first glance, the inequality looks somewhat surprising, since

$$(fg)'' = f''g + fg'' + 2f'g'.$$

Thus, one would naturally expect a few additional terms on the right-hand side, rather than a formula resembling the Leibniz formula of first-order differential calculus. That is, we simply obtain

$$\|(fg)''\|_2 \leq \|f\|_\infty \|g''\|_2 + \|g\|_\infty \|f''\|_2 + 2\|f'g'\|_2.$$

By Hölder's inequality, $\|f'g'\|_2 \leq \|f'\|_4 \|g'\|_4$. Moreover,

$$\int_0^\pi f'^4 dx = -f f'^3 \Big|_0^\pi - 3 \int_0^\pi f f'^2 f'' dx = -3 \int_0^\pi f f'^2 f'' dx$$

since $f(0) = f(\pi) = 0$. Now

$$\int_0^\pi |f f'^2 f''| dx \leq \|f\|_\infty \|f'^2\|_2 \|f''\|_2 = \|f\|_\infty \|f'\|_4^2 \|f''\|_2.$$

Thus

$$\|f'\|_4 \leq \sqrt{3} \|f\|_\infty^{1/2} \|f''\|_2^{1/2},$$

a particular form of the Gagliardo-Nirenberg inequality. To complete our reasoning, we use the geometric-arithmetic inequality

$$\begin{aligned} 2\|f'\|_4 \|g'\|_4 &\leq 6\|f\|_\infty^{1/2} \|f''\|_2^{1/2} \|g\|_\infty^{1/2} \|g''\|_2^{1/2} \\ &\leq 3(\|f\|_\infty \|g''\|_2 + \|g\|_\infty \|f''\|_2). \end{aligned}$$

Then

$$\|(fg)''\|_2 \leq 4(\|f\|_\infty \|g''\|_2 + \|g\|_\infty \|f''\|_2).$$

Surprisingly, note that the Leibniz inequality with constant 1 holds with pure sine modes. Indeed, considering the modes $f(x) = \sin kx$ and $g(x) = \sin mx$, we obtain

$$\sin kx \sin mx = \frac{1}{2} (\cos(k-m)x - \cos(k+m)x).$$

A direct calculation shows that

$$(fg)'' = \frac{1}{2} \left(-(k-m)^2 \cos(k-m)x + (k+m)^2 \cos(k+m)x \right).$$

By the orthogonality of the cos functions in $L^2(0, \pi)$, we have

$$\|(fg)''\|_2 = \frac{\sqrt{\pi}}{2\sqrt{2}} \sqrt{(k-m)^4 + (k+m)^4}.$$

Moreover, one finds that

$$\|f''\|_2 = \sqrt{\frac{\pi}{2}} k^2 \quad \text{and} \quad \|g''\|_2 = \sqrt{\frac{\pi}{2}} m^2.$$

And a simple algebraic computation yields

$$\frac{1}{2} \sqrt{(k-m)^4 + (k+m)^4} \leq k^2 + m^2.$$

Indeed, we take the square of both sides and obtain $0 \leq (k^2 - m^2)^2$. Hence, the equality occurs above if $f = g = \sin kx$, and the inequality also follows.

To find a counterexample to the inequality, a natural approach would be to take a first-order variation of the equality case. Let us define

$$f = g = \sin x + \varepsilon \sin 2x.$$

For a sufficiently small $\varepsilon > 0$, a numerical computation suggests the failure of the Leibniz inequality, and

$$\|(fg)''\|_2 > \|f\|_\infty \|g''\|_2 + \|g\|_\infty \|f''\|_2$$

holds. Indeed, with $\varepsilon = 0.1$, we have

$$\frac{\|(fg)''\|_2}{\|f\|_\infty \|g''\|_2 + \|g\|_\infty \|f''\|_2} \approx 1.0007730598.$$